in India in 2009. PyCon India was probably one of the first language-specific tech conferences in India, and it has since then grown to be one of the largest PyCons in the world. This year, due to the coronavirus situation, we're conducting the conference online. I'm also an active contributor to the Stack Overflow website, where I rank among the top 0.29 per cent worldwide for giving answers to questions."

The lack of discipline is a challenge

Ibrahim feels that a lot of people don't seem to realise that contributing something significant to a project requires a large amount of work. The main challenge is to develop patience and dedication to spend enough time to understand a project so that one can contribute to it. There are smaller problems, like some projects do not have enough contributors to help with technical problems, but overall, the main problem is the lack of discipline to put in the time necessary to achieve some level of proficiency.

He adds, "Mostly, it is just a matter of putting in the work and having the patience to find the people who can help you with it. I remember when I was using a test framework called py.test for the first time, I had some questions about writing a plugin for it. I found a mailing list that had people who worked in the area and sent a few emails there. The creator of the tool, Holger Krekel, responded to my email and solved my problem. One shouldn't be ashamed of asking questions and must have the discipline to put in the work necessary to move forward."

Knowledge and skills are needed to thrive in the new ecosystem

He further says, "During my work and interactions with students, I noticed a toxic mix of thirst for knowledge and complete lack of guidance at engineering colleges. Many of the professors teaching at tier-2 and tier-3 engineering colleges (and there are a large number of these catering to students) don't have any connection with the industry and the real world. This handicaps the students when they come out of college, and they're forced to take courses from small institutes of questionable repute, further damaging their career prospects. We want to change that."

He feels he can fix this problem at the root. Genskill aims to give students the knowledge and skills to thrive in the new knowledge ecosystem, and guide them while they're doing their engineering to reach their full potential.

He says, "Imagine a student coming out with a solid portfolio of serious projects and technical work, the ability to collaborate and communicate well, and very confident in her/his abilities… how solid his/her career is going to be. Then imagine an army of engineers coming out that companies can hire… how solid the technology industry in India will be. That's my vision, and that's what I am working towards right now."

For students, the platform provides content, guided courses, assessment, and evaluation modules that develop individual and collaborative skills required by the job market. Colleges can monitor a student's progress and provide personalised training that will increase student recruitment. This will help them improve student enrolment and increase college revenues. Companies can make data-driven hiring decisions, get access to a wider talent pool, and reduce fresher training costs. The offering focuses on individual and collaborative learning through technology and community building. By using an app, it will keep students engaged and continuously upskill them while monitoring their progress and areas of competence. It will also create and nurture student communities at the ground level to create a culture of acquiring job skills.

Enjoy the ride, it's not an exam

Ibrahim's advice to students and freshers starting out in this field is: "There are a few basics like learning a programming language and some basic tools like Git to contribute to projects. This is not hard to do and can easily be done within a few weeks of part-time work. What's challenging is the ability to put in practice hours and improve your skill to a level that becomes useful for people."

He lists a few points that young people should focus on while starting out:

1. Learn the tools of the trade (developer tools, languages, etc).
2. Practice your skills with the problem sets on sites like hackerrank, exercism.io, etc.
3. While you're doing that, start making small contributions to projects on GitHub. Most projects have open bugs that have been marked as contributor-friendly. Make small contributions, solicit feedback, and improve your skills.
4. Also, contribute in non-code ways. For example, answer questions on Stack Overflow, hang out on fora where the users and developers hang out, and help out with questions.
5. Try to network with like-minded people in the areas you're working. For all the havoc that the coronavirus has inflicted on our society, one of the arguably good things is that many tech conferences have moved online. This opens them to even student audiences who could otherwise not afford them. Take advantage of this and build your circle.
6. Finally, enjoy the ride. This is not an exam or a set of things to do to get a job. It's a process and a journey that will continuously benefit you.

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